ANNEXURE I

FORMAT - TECHNICAL SPECIFICATION FOR PURCHASE OF REVERSE OSMOSIS PLANT (2000 litre / hour) WITH ACCESSORIES

***General features*:**

i) The system shall be completely assembled, pre-piped, pre-wired and preferably skid mounted.

ii) The system shall have microprocessor based control panel for ease of operation and inbuilt process logic.

iii) The system shall have auto start/ stop based on water level in the supply tank.

iv) The system shall be equipped with Electrical panel/ MCB for plant protection and shall have integrated raw water pump with the unit.

v) The system shall have integrated pretreatment modules for removal of suspended solids along with pressure gauge.

vi) The system shall have inbuilt protection to high pressure pump by way of low/high pressure switch.

vii) The system shall have the membrane of General Electric (GE) or Hydranutics make. Glycerin filled Stainless Steel pressure gauge shall be provided for durability. Online rotameter shall be provided for measuring flow. Blending cartridge shall be provided to adjust taste/TDS and followed by UV disinfection for total safety. System should have automatic backwashing of filters.

viii) The system shall be capable of working 24hrs.x365 days by using the water provided through Borewell / Municipal water supply.

***Technical features*:**

Note: *Kindly adhere to the technical specification of your firm as per the technical specifications mentioned in Annexure I. If you are adhering to the technical specifications mentioned in annexure I, then kindly mention as "as per requirement" in column III. If any deviation or made, mention the deviations very clearly in column III.*

Sr. No.	Description (Column I)	Required specification (Column II)		Mention your firm specifications without deviations for all parameters. Mention specifically if any deviations made (Column III)
	RO Plant parts and its Specifications (2000 litres/hour)			
1.	**Raw Water Feed Pump**	**Type**	**Centrifugal**	
		Flow rate	3000 to 6000 Litres/Hour	
		Discharge Capacity	2.5 kg/cm2 (min.)	
		Motor Rating	2.0 HP (min.)	
		Electricity	415 V, Three phase	
		Quantity	1No	
		Make	KBL, KSB, Crompton, Shakti or equivalent	
2.	**Dual Media Filter (DMF)**	Capacity (min.)	3000 to 6000 Litres/Hour	
		Operating Pressure (Min)	3.5 kg/cm2	
		Type of Valve	Single- Multiport	

		Media	Anthracite + Support Bed.	
		Bed depth (min.)	800mm	
		Quantity	1No	
		Vessel Make	Pentair, wave or Equivalent	
3.	**Activated Carbon Filter (ACF)**	Capacity	3000 to 6000 Litres/Hour	
		Min. Operating Pressure	1.5 kg/cm2	
		Media	Activated Carbon + sand	
		Quantity	1No	
		Vessel Make	Pentair, wave Ion Exchange or Equivalent	
4.	**Antiscalant Dosing System**	Dosing Pump		
		Type	Electro-magnetic	
		Capacity	1.5 Litres/Hour	
		Discharge Pressure	4 kg/cm2	
		Quantity	1No	
		Make	E dose, hydra- cell, Sandur or Equivalent	
		Dosing Tank		
		Capacity	100 Litres	
		Material of Const.	HDPE	
		Quantity	1No	
		Make	Sintex, WimPlast or Equivalent	

5.	**Micron Cartridge Filter (For removing suspended particles >5 micron)**	Capacity	3000 to 6000Litres/Hour	
		Material of Cartridge	Polypropylene	
		Filter Rating	5 micron	
		Length	20”	
		Quantity	2 Nos.	
		Make	Pratham, Aquapuro or Equivalent	
6.	**High Pressure Pump**	Type of Pump	Vertical Multistage centrifugal	
		Capacity	3000 to 6000 Litres/Hour	
		Discharge Pressure (Max.)	12 - 15 kg/cm2	
		Material of Construction	SS 304	
		Electricity	415 V, Three phase	
		Quantity	1	
		Make	Nanfang, CRI, Yuken or Equivalent	
7.	**Reverse Osmosis Membrane**	Permeate capacity	2000 Litres/Hour	
		Membrane Type	Spiral wound TFC- polyamide	
		Size	Dia. 4’’ X 40’’ long	
		No. of Membrane	9	
		No. of Membrane Housing	3	
		Skid Material of Construction	Mild Steel – Powder Coated	
		Make of Membrane	Gen. Electric, Hydronautics or Equivalent	

8.	**UV system**	Capacity	2000Litres/Hour	
		Quantity	1	
		Make	Sukrut or Equivalent	
9.	**Instrument List**	A) Pressure gauges	5 Nos.	
		Range	0 – 21 bar	
		Dial Size	4 inches	
		Type	Bourdon	
		B) Pressure switches	2 Nos.	
		Range	0-450 psi	
		C)Conductivity meter	1 (Panel Mounted)	
		D) Rotameter	2	
		E) Level switch	2	
		F) Digital TDS Meter	1	
10.	**Electrical Control Panel**	A) Starters, overload relays, single/three phase controller for pump	Should be provided with adequate capacity Havells/Ancho r/ Equivalent	
		B) Voltmeter, Ammeter, MCB's	Should be provided with adequate capacity	
		C)PLC & Touch Panel	Should be provided	
		D) Feed/ Product water conductivity meter	Should be provided	
11.	**Raw water tank**	Capacity	10000 litre	
		Make	Sintex triple layer white	

12.	**Purified water tank**	Capacity	5000 litre	
		Make	Sintex triple layer white	

Electricals

- MCC Panel board With power cables and control cables
- Panel board, switches and cables – make and wire gauge should be mentioned.
- Electrification Material like Cables, Cable trays, Clamps, Earthing, G.I strips necessary for power wiring from Panel to respective Motors.
- Earthing, consisting of earth pits, running of earth wire etc for MCC panel and individual motor.

Supply, installation, testing, Commissioning and trial run

- Complete erection of machines, drives, structure and electrical till MCC panel.
- Commissioning of total system to production should be done satisfactorily.
- Annual Maintenance charges should be free at least for first one year from the date of commissioning of Reverse Osmosis plant.
- Any service work should be attended within 48 hours.
- Guarantee and Warranty period should be specified.
- Spare parts with price list should be submitted.

Other terms and conditions:

- Company should have at least three years of experience.
- One year guarantee must be provided for the plant.
- Tender with incomplete information or wrong information will be summarily rejected.
- Horse power details for all motors should be mentioned correctly.
- Material of Construction (MOC) should be mentioned for all materials.

- Adequate spares required for one year should be mentioned.
- Spares offered freely along with the reverse osmosis plant should be clearly mentioned.
- The decision of Tender Evaluation Committee is final for any discrepancies in GST rates for machineries and commissioning.